SOLUTIONS

Holy Cross College
Science Department
Year 9 Atom Section Test **Mark: /50**

HOLY CROSS COLLEGE

Part A: Multiple Choices [10 marks]

Full Name: ....

1. γ, β, α, B
IN ORDER, the above symbols mean:

A Gamma, Beta, Alpha, Boron
B Gamma, Boron, Alpha, Beta
C Alpha, Beta, Gamma, Beryllium
D Alpha, Beryllium, Gamma, Beta

2. The nucleus of which atom is identical to an alpha particle?

A H - Hydrogen
B He - Helium
C Li - Lithium
D Ti - Titanium

3. Which of the following statements about atoms is true?

A An atom is mostly empty space.
B Neutrons and protons are found in the nucleus with the electrons.
C Electrons circle the nucleus all the time.
D The number of neutrons found in the nucleus of an atom is always the same.

4. The mass number of an element is the total number of:

A electrons in an atom
B protons in an atom
C protons and electrons in an atom
D protons and neutrons in an atom

5. The atomic number of an element is the total number of:

A electrons in an atom
B protons in an atom
C protons and electrons in an atom
D protons and neutrons in an atom

6. An atom contains 4 protons, 5 neutrons and 4 electrons. Its mass number is:

A 4
B 5
C 9
D 13

7. An atom contains 4 protons, 5 neutrons and 4 electrons. Its atomic number is:

A 4
B 5
C 9
D 13

8. An ion is:

A an atom that has lost or gained neutrons
B an atom that has lost or gained protons
C an atom that has lost or gained electrons
D one or more atoms that have lost or gained electrons

9. A charged atom is called:

A an isotope
B an ion
C a proton
D radioactive

10. Define ionising radiation:

A Radiation from the ionosphere.
B Radiation produced when atoms are ionised.
C Any type of electromagnetic radiation.
D Radiation that can remove electrons from atoms and molecules.

Part B: Short Questions [35 marks]

1. What makes the nuclei of some atoms naturally unstable? [1]

• Protons and neutrons not held together strongly.

• Nucleus has too much energy.

[Either answer - 1 mark]

2. How did Rutherford's nuclear model differ from Thomson's plum pudding model of the atom? [2]

• Dense nucleus in the centre with electrons orbiting. (1)

• Electrons were mobile - not embedded. (1)

3. Explain the difference between the terms 'atomic number' and 'mass number'. [2]

ATOMIC NUMBER - number of protons (1)

MASS NUMBER - number of protons + neutrons. (1)

4. List which subatomic particles in a neutral atom have:

a) equal but opposite charges: [1]

proton + electron (1)

b) approximately the same mass: [1]

proton + neutron (1)

5. For each of the following, write symbol of the ion formed.

a) Magnesium atom (Mg) loses 2 electrons: [1]

Mg^{2+} (1)

b) Nitrogen atom (N) gains 3 electrons: [1]

N^{3-} (1)

6. The atomic number of uranium (U) is 92. The atomic number of thorium (Th) is 90. Uranium-238 undergoes alpha decay to form thorium-234 and an alpha particle, according to the following equation:

$${}^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\alpha$$

a) Clarify what the number 238 tells us about the uranium-238 atom. [1]

• Has 92 protons and 146 neutrons.

b) Describe the composition of an alpha particle. [1]

• He nucleus - 2 protons and 2 neutrons.

7. In a chemical reaction, will a sulfur atom gain or lose electrons to ionise? How many? Why is this?
Hint: look at your periodic table. [3]

• Gain (1)

• 2 electrons (1)

• Fills up its outer shell. (1)

8. Oxygen has three isotopes, as shown below. Describe the similarities and the differences between them.
Hint: compare the subatomic particles in each.

${}^{16}_{8}\mathrm{O}$ ${}^{17}_{8}\mathrm{O}$ ${}^{18}_{8}\mathrm{O}$

[3]

• Same number of protons. (1)

• Different number of neutrons. (1)

• Same element - same chemical characteristics. (1)

9. Nuclear radiation is known to be able to harm the human body, but is also able to be used safely if precautions are taken.

   a) List two major long-term effects on humans caused by cell damage resulting from exposure to large amounts of nuclear radiation. [1]

   • Cancer

   • Genetic mutations

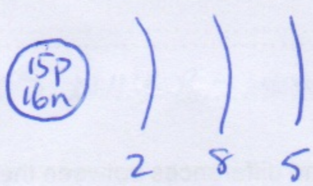
   b) Describe two uses of nuclear radiation in medicine and/or industry. [2]

   ANY 2 WITH REASONABLE SHORT DESCRIPTION – 1 mark each.

10. Draw the following atoms with the correct electron shell configurations. Label the following parts: nucleus, neutron, proton, electron, electron shell (6)

   a) phosphorus (atomic number 15): [3]

$^{31}_{15}P$



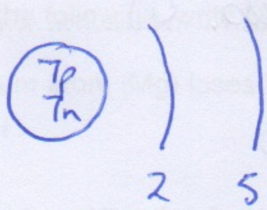
Correct element – 1 mark

Correct nucleus – 1 mark

Correct electron shells – 1 mark

   b) an atom with atomic number 7: [3]

$^{14}_{7}N$



11. The atoms below are written using the letters V to Z instead of their correct chemical symbols.

$${}^{40}_{18}V \quad {}^{20}_{10}W \quad {}^{40}_{19}X \quad {}^{39}_{19}Y \quad {}^{40}_{20}Z$$

Use the letters to identify the following:

a) Which two atoms have the same number of neutrons? [1]

${}^{39}_{19}Y$ ${}^{40}_{20}Z$

b) Which atom has the smallest mass number? [1]

${}^{20}_{10}W$

c) Which two atoms belong to the same element? [1]

${}^{40}_{19}X$ ${}^{39}_{19}Y$

d) Which atom has the electron configuration 2, 8? [1]

${}^{20}_{10}W$

12. Answer true or false for each statement. [5]

i) Isotopes have different mass numbers T / F

ii) Carbon-12 has six neutrons. T / F

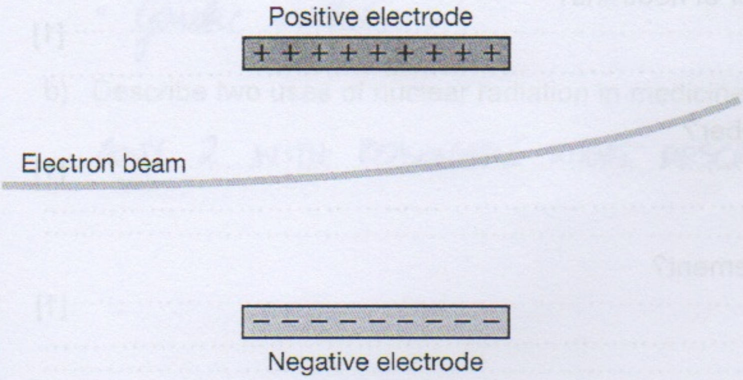
iii) Neutrons have a negative charge. T / F

iv) Alpha particles pose a great external danger to the body. T / F

v) Beta particles travel about 100 cm through the air. T / F

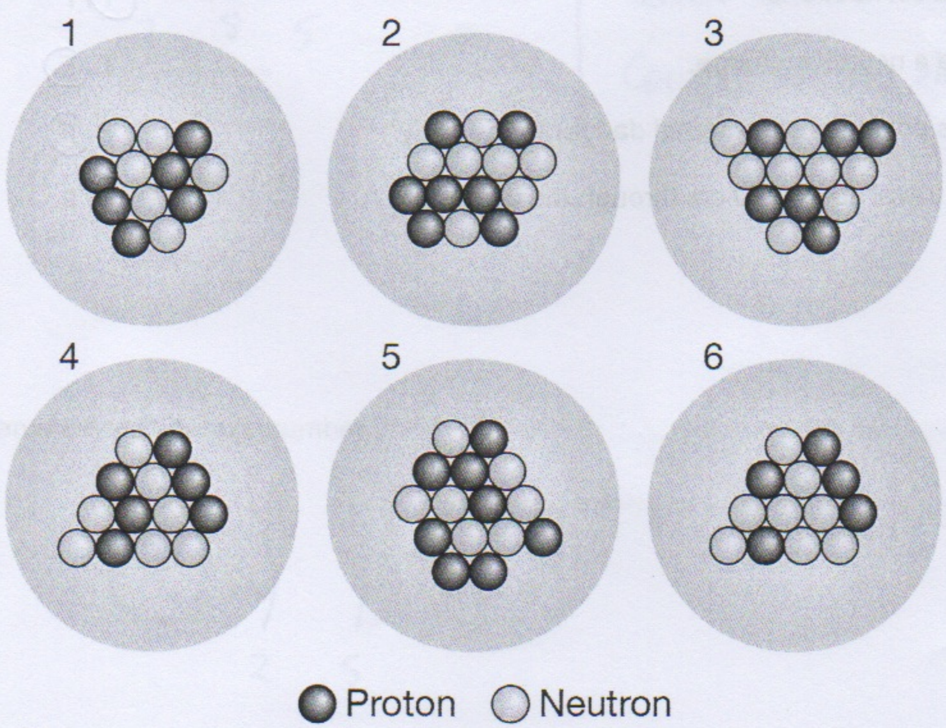
Part C: Comprehension [5 marks]

1. To learn more about electrons, Joseph John Thomson passed an electron beam through a pair of electrodes. He found that the electron beam shifted towards the positive electrode. Given that like charges repel each other and opposite charges attract, deduce what Thomson could conclude about the charge of the electrons.



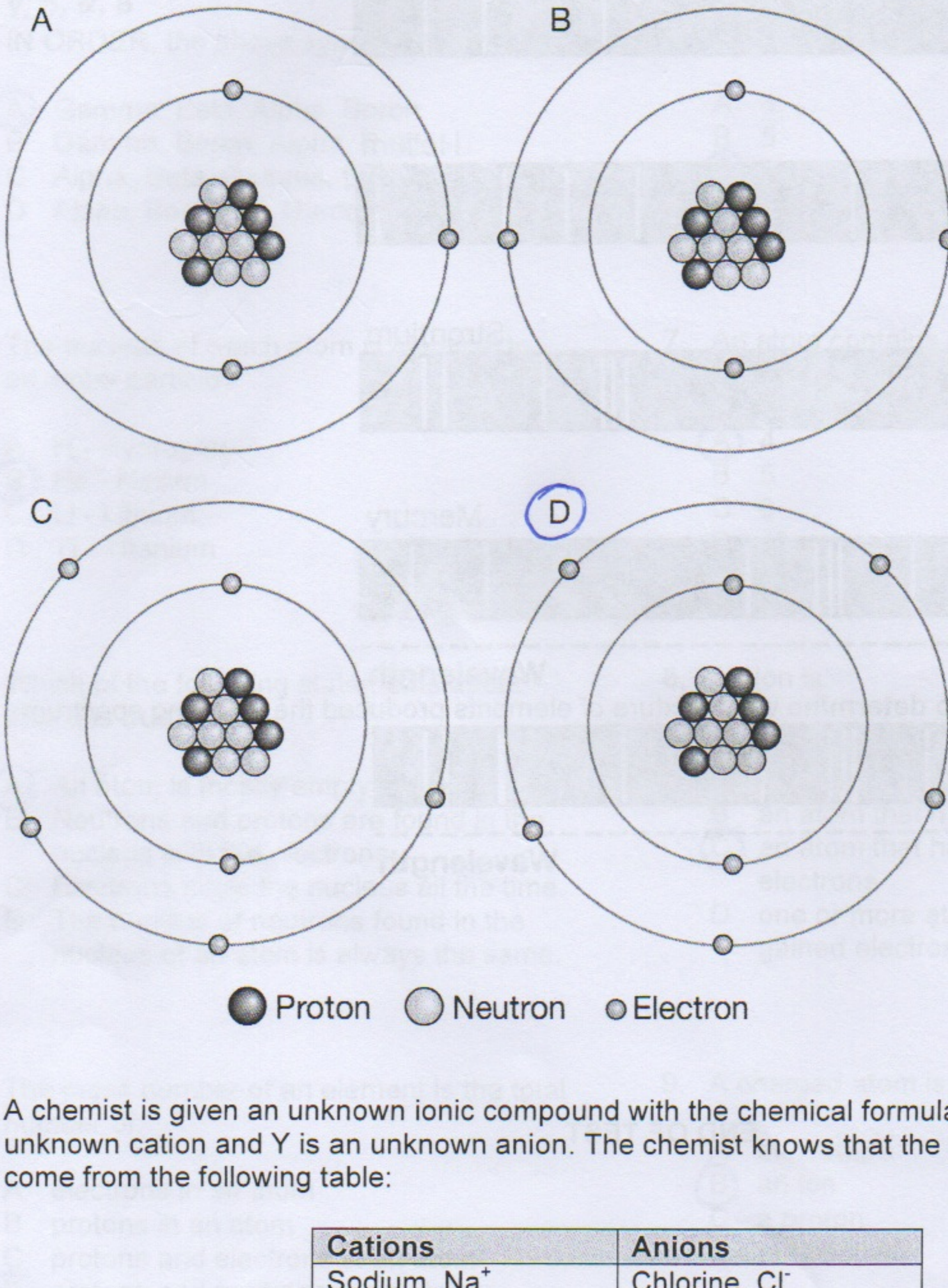
A Electrons are charged, but might be positively or negatively charged.
B Electrons are negatively charged.
C Electrons are positively charged.
D Nothing can be concluded about the charge of the electrons.

2. Isotopes are atoms that have the same atomic number but a different mass number. In other words, they have the same number of protons but a different number of neutrons. Determine which of the following atoms are isotopes of the same element.



A 1 and 4
B 1, 3 and 4
C 2 and 5
D 3 and 6

3. Atoms are always electrically neutral because they have the same number of protons and electrons.
Negative ions (**anions**) form when an atom gains electrons. They have a negative charge equal to the number of electrons gained.
Positive ions (**cations**) form when an atom loses electrons. They have a positive charge equal to the number of electrons lost.
Use this information to determine which of the following images represents an ion with a charge of -2.



4. A chemist is given an unknown ionic compound with the chemical formula X_2Y_3, where X is an unknown cation and Y is an unknown anion. The chemist knows that the cation and anion must come from the following table:

Cations	**Anions**
Sodium, Na^+	Chlorine, Cl^-
Magnesium, Mg^{2+}	Oxide, O^{2-}
Aluminium, Al^{3+}	Nitride, N^{3-}
Chromium (IV), Cr^{4+}	Carbide, C^{4-}

From this table, deduce the chemical name of the unknown ionic compound.
A aluminium oxide
B chromium (IV) nitride
C magnesium nitride
D aluminium carbide

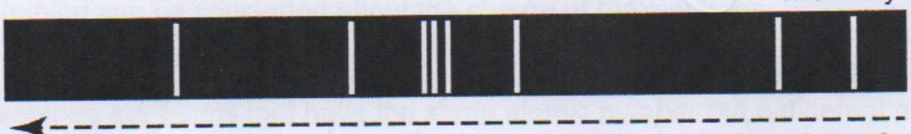
5. Atoms heated in a Bunsen burner produce coloured light as the electrons jump between electron shells. When an atom's coloured light is put through a spectrum analyser, it is seen to be made up of several different colours of light. This is the atomic spectrum, which is like a unique fingerprint for every type of atom.
Here are the atomic spectra for hydrogen, helium, strontium and mercury:

Hydrogen

Helium

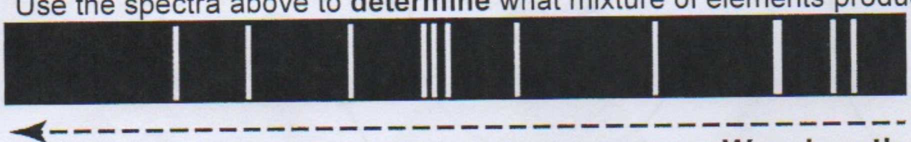
Strontium

Mercury



Wavelength

Use the spectra above to **determine** what mixture of elements produced the following spectrum:



Wavelength

A hydrogen and helium
B hydrogen and mercury
C helium and mercury
D hydrogen and strontium

END OF TEST